

SMART INDIA HACKATHON 2026

PIXEL-Srishti

An Agentic Geospatial AI Assistant for Multi-Sensor Satellite Image Analysis and Watershed Development Monitoring

Field	Details
Team Name	JALDI THE LATE
Theme	Space Technology & Miscellaneous
PS Category	Software
Problem Statements Addressed	1) Geospatial Visualization & Interpretation of Watershed Development using SRISHTI-DRISHTI Satellite Data. 2) SatQuery AI: Agentic Vision-Language Assistant for Single & Multi-Image Remote-Sensing Analysis (ISRO/SAC).
Organisation	Ministry of Rural Development & ISRO–Space Applications Centre (SAC)

1. Team & Mentor Details

S.No	Name	Role	Registration Number
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6	Abhik Kumar	Data Engineer: Geo-data Pipeline	25BAI0059
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2. Problem Statement Description

Right now, satellite image analysis tools are all built to do just one job each, one tool can classify land types, another can spot changes over time, another can answer basic questions about an image, but none of them talk to each other, and using any of them requires specialized GIS training most people don't have.

On top of that, satellites capture images in different ways: regular photos (optical imagery), detailed color-band data (multispectral data), and radar images that can see through clouds and even trees (SAR: Synthetic Aperture Radar). Each type shows something different and useful, but right now there's no single tool that lets an ordinary person compare or combine them just by typing a question in plain English, like "what changed here in the last year?"

This gap causes real problems on the ground. Organizations like ISRO collect huge amounts of high-quality satellite data (through platforms like SRISHTI-DRISHTI), and field teams separately collect thousands of on-ground photos and survey notes during projects like rural development and land/water resource management. But all this data just sits there, nobody has an easy, automated way to turn it into clear, usable insights about things like land use, drainage, plant cover, soil wetness, or water bodies changing over time. So despite having tons of data, decision-makers still don't get fast, clear answers when they need them.

Keywords: *SatQuery AI, Agentic Vision-Language Assistant, Remote Sensing AI, SRISHTI-DRISHTI Platform (ISRO/SAC), Multimodal Analysis, Optical-SAR Pairs, Multitemporal Series, Visual Question Answering (VQA), Watershed Development, Resource Management, Geo-Tagged Photographs, 30 m Resolution Data, Natural Language Queries, GIS-Free Decision Making.*

2.1 Convergence Rationale

The problem statement converges on the same underlying gap: the absence of an intelligent, query-driven interpretation layer between raw satellite/geo-coded imagery and the field-level decisions that depend on it. SRISHTI-SENTINEL treats watershed monitoring as the flagship real-world application of a general-purpose agentic remote-sensing assistant solving the domain-specific watershed problem while building the reusable, multi-task AI backbone the SatQuery AI statement calls for.

3. Solution Objectives

- Build an agentic vision-language system that answers natural-language queries over single remote-sensing images, co-registered optical–SAR pairs, and bi-temporal image pairs.
- Fine-tune remote-sensing-adapted models (not a generic LLM/VLM) using BigEarthNet, VRSBench, RSVQA, and CDVQA as prescribed, so outputs are domain-grounded and auditable.
- Apply this backbone to watershed development monitoring: automatically generate land-use, drainage, vegetation, and water-body thematic maps from SRISHTI-DRISHTI 30 m imagery and geo-coded field photographs (potentially can be expanded further as well).
- Detect and describe spatial change over time (e.g, new water-conservation structures, vegetation recovery, land degradation) to support evidence-based watershed planning.
- Provide an orchestration layer that automatically selects the right specialist model/tool per query, returns visual evidence and confidence scores, and produces a downloadable, decision-ready report for planners and administrators.
- Keep the system scalable and low-cost so it can be replicated across watershed programs and other remote-sensing use cases beyond the pilot region.

4. Methodology

4.1 System Architecture

SRISHTI-SENTINEL is structured in four layers:

- (i) a data-ingestion layer that validates and pre-processes GeoTIFF/TIFF optical, multispectral, and SAR inputs along with SRISHTI-DRISHTI tiles
- (ii) a specialist-model layer of fine-tuned remote-sensing models for distinct tasks
- (iii) an agentic controller that interprets each natural-language query, checks input compatibility, and routes it to the correct specialist model(s)
- (iv) a presentation layer, an interactive web GUI, that returns thematic maps, change overlays, textual answers, confidence scores, and an auditable execution summary.

4.2 Machine Learning Models

Model	Role in the Pipeline
Remote-Sensing VQA / Captioning Model	A vision-language backbone (BLIP-2 / LLaVA-style architecture) fine-tuned on BigEarthNet + VRSBench for scene description and visual question answering on single optical/SAR images.
Text-Guided Grounding Model	A grounding model (Grounding DINO-style, RS-adapted) trained on VRSBench to localise objects/regions referenced in a query, e.g. 'highlight the water body'.

Bi-Temporal Change Model	A Siamese-encoder change-detection network (ChangeFormer / U-Net-based Siamese architecture) evaluated on CDVQA, producing both a change description and an optional pixel-level change mask.
Optical–SAR Fusion Model	A dual-branch fusion CNN that combines optical/multispectral and SAR feature maps for joint built-up and water-body extraction, exploiting SAR's cloud-penetration advantage.
Watershed Thematic Segmentation Model	A U-Net / DeepLabV3+ semantic segmentation model fine-tuned on SRISHTI-DRISHTI 30 m tiles to classify land use, drainage lines, vegetation density, and water-conservation structures.
Agentic Orchestrator	An LLM-based controller (open-source instruction-tuned model, function-calling / tool-use pattern) that classifies query intent, checks image modality & metadata, sequences specialist models, and merges outputs into one evidence-grounded response.
Trend Forecasting (extension)	An LSTM/ARIMA time-series module over historical change outputs to flag watersheds at risk of degradation before it becomes visually severe.

4.3 Technology Stack

- Backend / ML: Python, PyTorch, HuggingFace Transformers, GDAL/Rasterio for geospatial I/O
- Agentic orchestration: LangGraph-style tool-calling agent over the specialist model registry
- Geospatial processing: QGIS/GDAL pipelines, GeoTIFF handling, SRTM/Sentinel-derived auxiliary layers
- Frontend: React-based web GUI with an interactive map viewer (Leaflet/Mapbox) for thematic and change overlays
- Infrastructure: Cloud GPU for training/inference, containerised deployment (Docker) for portability to state/district servers

4.4 Datasets

- BigEarthNet — remote-sensing adaptation/domain fine-tuning of the vision-language backbone
- VRSBench, RSVQA — evaluation of single-image captioning, grounding, and VQA
- CDVQA — evaluation of multi-temporal change description and change-based VQA
- SRISHTI-DRISHTI 30 m platform + Cartosat-2S / RISAT pairs — India-specific watershed application and ISRO/SAC evaluation set

5. Feasibility & Viability

5.1 Key Risks & Mitigation

Risk	Mitigation
Limited GPU time for fine-tuning multiple specialist models	Use parameter-efficient fine-tuning (LoRA/QLoRA) on pre-trained backbones instead of full fine-tuning

Scarcity of labelled watershed-specific ground truth	Bootstrap labels from existing geo-tagged field photographs plus weak supervision from SRISHTI-DRISHTI thematic layers
Agentic controller mis-routing complex multi-modal queries	Constrain the tool registry with strict input-compatibility checks before execution; log an auditable execution trace for every query
Cloud cover affecting optical imagery	SAR fusion pathway is designed specifically to provide all-weather, day-and-night coverage as a fallback

6. Expected Outcomes & Impact

- A working web application where a planner can upload or select imagery and ask natural-language questions to get instant, evidence-backed watershed insights, replacing weeks of manual field-survey interpretation.
- Automated thematic maps (land use, drainage, vegetation, water-conservation structures) and change-detection reports generated directly from SRISHTI-DRISHTI data, improving monitoring accuracy and turnaround time.
- A reusable, general-purpose remote-sensing agentic backbone that can be redeployed for disaster management, forest monitoring, urban planning, or agricultural assessment beyond the watershed pilot — directly satisfying the scalability goal named in the SatQuery AI problem statement.
- Stronger evidence base for watershed policy and fund allocation decisions, and reduced dependency on scarce GIS expertise at the field level.
- Improved transparency: every answer ships with an auditable execution trace (models used, confidence, visual evidence), making outputs suitable for administrative and audit use.

7. Budget Estimate

The following is an indicative budget for building and demonstrating a working prototype through to grand-finale readiness. Figures assume use of free-tier/open-source tooling wherever possible, with paid cloud GPU time as the primary cost driver.

Component	Description	Est. Cost (INR)
GPU Compute (Cloud)	Fine-tuning VLM/change-detection models on BigEarthNet, VRSBench, CDVQA (approx. 300 GPU-hrs, A100-class)	45,000
Satellite & Reference Data	SRISHTI-DRISHTI 30m imagery access, Cartosat-2S / RISAT sample sets, ground-truth labels	10,000
Storage & Hosting	Cloud object storage for GeoTIFF datasets + web-app hosting (demo deployment)	8,000
Software & APIs	Open-source frameworks (no license cost); optional mapping/geocoding API credits	5,000

Domain Expert Consultation	Watershed/agronomy and GIS expert review sessions for validation	12,000
Hardware Peripherals	Edge test device / field data-collection kit for demo (optional)	15,000
Miscellaneous & Contingency	Printing, travel for grand finale, buffer (~10%)	10,000
Total (approx.)	<i>Estimated for prototype + finale-readiness stage</i>	₹1,05,000

8. Potential Scope & Future Work

- Extend the agentic framework to additional remote-sensing tasks (crop-yield estimation, flood-extent mapping, urban-sprawl tracking) by adding new specialist models to the same registry without redesigning the controller.
- Integrate with state/district Watershed Management Information Systems for direct deployment by rural development departments.
- Add a mobile-first, low-bandwidth interface so field engineers can query imagery on-site with limited connectivity.
- Incorporate IoT soil-moisture and rain-gauge feeds to fuse ground-sensor data with satellite-derived insight for tighter-loop watershed decisions.
- Open the model registry as an API so other ministries/departments can plug in their own fine-tuned specialist models.

9. References

- SRISHTI-DRISHTI satellite data platform (30 m spatial resolution)
- BigEarthNet — large-scale Sentinel benchmark archive for remote-sensing deep learning
- VRSBench, RSVQA — remote-sensing visual question answering and captioning benchmarks
- CDVQA — change-detection-based visual question answering benchmark
- Cartosat-2S (optical) and RISAT (SAR) — ISRO satellite missions referenced for the evaluation dataset